Poisson Effect-Assisted Replication Lithography and Its Implementation in a Roll-to-Roll Continuous Fabrication System

Minsu Kim, Han Jun Park, Moon Kyu Kwak*
Kyungpook National University
E-mail: mkkwak@knu.ac.kr

포아송 효과 활용 복제 리소그래피의 응용과 롤투롤 기반 연속 생산 공정 적용

김민수, 박한준, 곽문규*,
Kyungpook National University
E-mail: mkkwak@knu.ac.kr

Abstract

Scalable fabrication of complex three-dimensional (3D) microstructures remains challenging: conventional imprint/soft-lithography replication often fails at demolding undercuts and closed-loop geometries, whereas two-photon polymerization (2PP) is intrinsically serial and low-throughput. Poisson Effect-Assisted Replication Lithography (PEARL) addresses this gap via a soft-lithography-compatible workflow that adds controlled compression to harness Poisson effect-driven lateral deformation of an elastomeric mold. Compression assembles split cavity parts into a closed 3D cavity for curing, and unloading reopens the cavity for fracture-free release. This work summarizes the PEARL process and design window, demonstrates replication of representative closed-loop 3D microstructures, and introduces a finite element method (FEM)-based iterative compensation method for master-mold design. We further apply PEARL to replicate biodegradable magnetic helical microswimmers and demonstrate a roll-to-roll (R2R) PEARL system for continuous replication on flexible substrates, positioning PEARL as a practical manufacturing route for 3D microdevices and microrobotic components.

Keywords: Poisson effect, soft lithography, roll-to-roll, magnetic microswimmers

1. Introduction

High-throughput manufacturing of 3D microstructures is increasingly required for microdevices, metamaterials, and microrobotic actuators, yet scalable production remains limited. Imprint-based replication is compatible with large-area and continuous processing; however, demolding-induced fractures and defects typically confine it to 2D/2.5D patterns, particularly for overhangs, undercuts, and closed-loop geometries. 2PP offers geometric freedom but is inherently serial, resulting in a strict productivity limitation. Here we introduce PEARL, which exploits Poisson effect-driven deformation of an elastomeric mold, and present basic replication of closed-loop 3D microstructures. We further demonstrate PEARL-enabled fabrication of helical microswimmers, and implementation of PEARL in a R2R continuous manufacturing system.

2. PEARL principle and baseline demonstrations

2.1 Working principle

PEARL uses an elastomeric soft mold (e.g., PDMS) whose negative cavity is intentionally split across facing walls separated by designed gaps, as shown in Fig. 1. Under applied normal compression, the elastomer expands laterally via the Poisson effect, closing the gaps and assembling the split cavity parts into a sealed 3D cavity. Replication resin placed between the mold and substrate is squeezed during deformation so that the assembled cavity remains filled; UV (or thermal) exposure polymerizes the resin in this “closed” state. After curing, unloading restores the mold and reopens the gaps, allowing the cured closed-loop structure to be released through the reopened gaps with minimal risk of fracture.

2.2 Basic demonstrations with closed-loop microstructures

Basic demonstrations using PDMS soft mold and UV-curable polyurethane acrylate establish a clear process window, as shown in Fig. 2(a-d). Insufficient strain yields incomplete gap closure and residual walls replicated from the open gap region. Excessive strain distorts the assembled cavity and can narrow or break features. Simple closed-loop microstructures verify that operating slightly above the closure strain typically improves sealing and resin squeeze-out while limiting distortion. These results motivate design rules that balance gap size, block geometry, and target strain to obtain repeatable assembly and release.

2.3 FEM-based iterative compensation for master mold design

For complex 3D shapes, fidelity is governed by the cavity geometry in the compressed, assembled state. Because PEARL operates at a relatively large strain range and soft mold materials are hyperelastic, simplified incompressible estimates are useful but insufficient for precise alignment. We therefore use nonlinear FEM analysis and an iterative compensation loop to define the target cavity in the closed state, simulate deformation at the target strain, quantify boundary error, update the master geometry in the opposite direction of the error, and iterate until the assembled cavity deviation is within tolerance.

3. Application to biodegradable magnetic helical microswimmers

PEARL is applied to replicate magnetically actuated helical

microswimmers, demonstrating scalable fabrication of functionally relevant 3D microrobots. A 2PP-printed master mold encodes split helix halves on opposing sides of vertical walls; a PDMS mold is cast; and PEARL compression assembles a complete helical cavity for UV crosslinking. Gelatin methacryloyl (GelMA) is used as a biodegradable hydrogel matrix, and iron-oxide nanoparticles (IONPs) are incorporated to enable magnetic actuation. Because PEARL relies on flood UV curing rather than focused laser writing, higher IONP loading can be processed by tuning exposure and viscosity; in this work, replication is demonstrated up to 30 wt% IONPs while preserving helical geometry.

Swimming tests under a rotating magnetic field (≈25 mT) show monotonic performance improvement with IONP loading. Maximum axial velocities of 113.7, 149.2, and 182.6 μm/s are achieved for 10, 20, and 30 wt% IONP, with corresponding step-out frequencies of 55, 100, and 110 Hz. GelMA microswimmers also degrade in collagenase-containing media, supporting transient operation and reducing residual burden after actuation in biomedical or biochemical environments.

4. Roll-to-roll PEARL for continuous replication

To bridge PEARL from parallel batch replication to continuous manufacturing, we developed a pilot scale R2R device to assess the R2R compatibility of PEARL, as shown in Fig. 3. To keep the hardware and operating conditions simple and readily controllable, the normal load required to compress the elastomeric soft mold was generated by adjusting the web tension of the feeding substrate, rather than by employing an additional pressurization module. This pilot R2R demonstration indicates that PEARL can be straightforwardly integrated into continuous production workflows, providing a practical route toward high-throughput manufacturing of 3D microstructures.

5. Conclusion

PEARL enables scalable replication of closed-loop 3D microstructures by assembling split cavities through Poisson effect-driven deformation during curing and disassembling them for defectless release. Basic demonstrations define the strain window, FEM-based iterative compensation improves master-mold accuracy under nonlinear deformation, and the microswimmer and R2R applicability demonstrations together outline a manufacturing pathway for functional 3D microdevices and microrobot components.

Acknowledgments

This work was supported by the National Research Foundation of Korea (Grant nos. RS-2022-NR070937, RS-2023-00250849, RS-2024-00462602, RS-2024-00432685).

References

1. M. Kim, D. Kyeong, M. K. Kwak, Poisson Effect-Assisted Replication Lithography for Rapid Fabrication of Three-Dimensional Microstructures, *Small Struct.* ***5***, 2300460 (Jun. 2024).

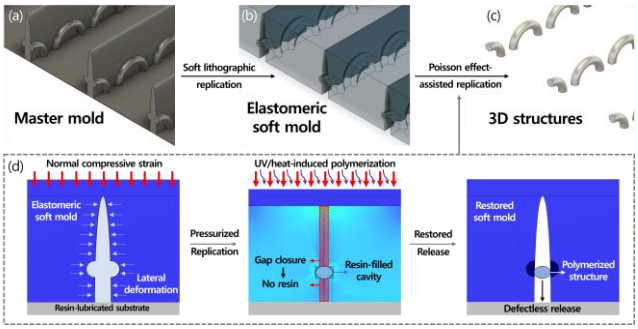

Fig. 1. Conceptual illustration of the PEARL process.

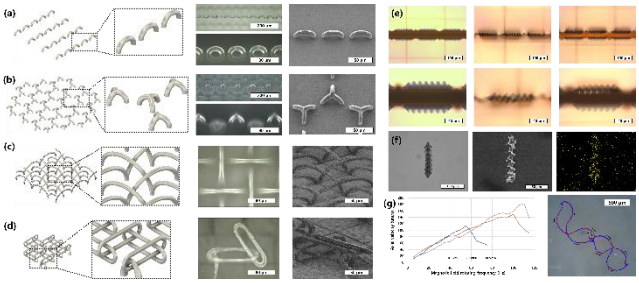

Fig. 2. Demonstrative examples of 3D microstructure replication via PEARL process.

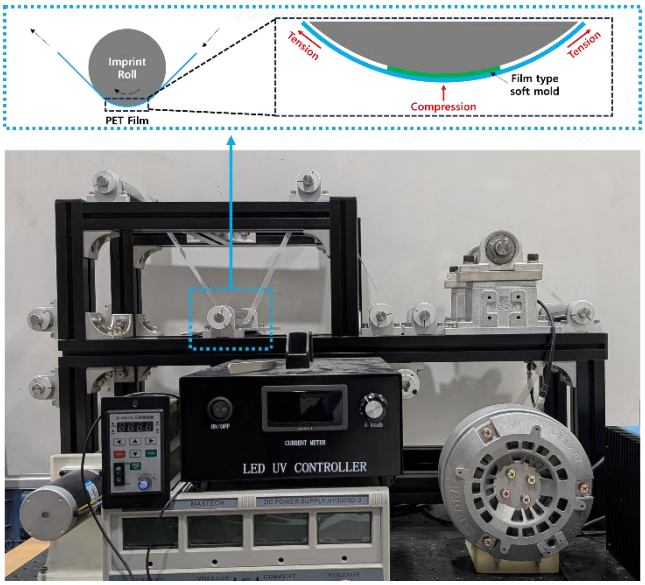

Fig. 3. Pilot roll-to-roll device for PEARL applicability demonstration.